

Self-Adaptive Model-Based Protection for IBR-Penetrated Networks: A Unified Five-Layer Framework

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Abstract—Inverter-based resources (IBRs) are reshaping fault current behaviour in distribution and sub-transmission networks: currents are bounded, current-limiting controllers activate within 2–5 ms, and the aggregate IBR penetration ratio k_{ibr} varies continuously over the life of the installation. Classical protection—overcurrent, distance, and differential—was designed for stiff synchronous machine fault currents and degrades measurably when $k_{\text{ibr}} > 0.25$. This paper presents the *Self-Adaptive Model-Based Protection System* (SAMBPS), a five-layer hierarchical architecture that replaces fixed relay settings with online model estimation, physics-constrained adaptation, and a stability-verified trip gate.

Layer 1 (L1) conditions raw CT/PT samples at 4 kHz and produces pre-fault phasor quantities. Layer 2 (L2) runs a Levenberg–Marquardt (LM) inverse zone estimator every scan cycle, outputting fault-zone parameters $\hat{\theta}$, the SVD-derived condition number κ_n , and the real-time IBR penetration estimate $\hat{k}_{\text{ibr}}$. Layer 3 (L3) updates the differential-slope threshold $k(t)$ and forgetting factor $\lambda(t)$ via projected-gradient adaptive laws whose constraint set $\mathcal{K}(\kappa_n)$ is gated by κ_n . Layer 4 (L4) evaluates a CUEP energy bound $V_{\text{cuep}}(k_{\text{ibr}})$ against the fault-cleared energy V_0 , providing a stability-based trip gate. Layer 5 (L5) issues trip/restrain decisions, enforcing a Stage-2 veto whenever $\kappa_n \geq 30$, and exchanges decisions wide-area via IEC 61850 GOOSE.

We unify sixteen theoretical contributions (C1–C16) and seven validation campaigns (G1–G7) under this architecture, demonstrating zone accuracy $\geq 95.5\%$ and false-alarm probability $P_{\text{FA}} \leq 1.6\%$ across 33 kV DSO field recordings, hardware-in-the-loop experiments on a SEL-411L relay, and a 1500-scenario IEEE 39-bus simulation study. The framework extends the valid observation window from 150 ms (sixth-order Park model) to 380 ms via a nine-state virtual synchronous machine (VSM) model that incorporates AVR and PSS dynamics.

Index Terms—adaptive protection, condition number, CUEP energy bound, differential protection, IBR penetration, inverter-based resources, Levenberg–Marquardt estimation, Lyapunov stability, microgrid protection, self-adaptive relaying, SVD interlacing, virtual synchronous machine, wide-area protection.

I. INTRODUCTION

The global transition to renewable generation is concentrating inverter-based resources (IBRs) — photovoltaic plants, DFIG wind turbines, battery storage, and virtual synchronous machines — in distribution and sub-transmission networks that were designed for radial, synchronous-machine-dominated fault currents. By 2030, projections indicate that the aggregate IBR penetration ratio $k_{\text{ibr}} = P_{\text{IBR}}/P_{\text{total}}$ will exceed 0.40 in most developed-country distribution systems [1], [2].

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A. Challenges for Classical Protection

Classical differential protection (87L, 87T, 87B) compares operate and bias currents against a fixed characteristic slope k :

$$I_{\text{diff}} \geq k \cdot I_{\text{bias}} + I_{\text{pickup}}.$$

This slope is set once during commissioning and assumes that fault currents will be dominated by synchronous machine contributions, which are large, stationary, and well-characterised. Under IBR penetration three failure modes emerge:

F1 — Bounded current: IBR current limiters clip output at 1.1–1.2 pu, reducing I_{diff} below the sensitivity threshold for remote high-impedance faults [3].

F2 — Non-stationary: The IBR control law transitions from current-controlled to voltage-controlled mode during sustained islanding, causing slow drift in the apparent slope $k^*(t)$ [4].

F3 — Shape factor: The bias current I_{bias} depends on the geometric mean of terminal currents; with mixed IBR/synchronous sources, the effective calibration factor β shifts by up to 18% [5].

B. Prior Art and Limitations

Adaptive protection schemes have been proposed based on rule-based setting groups [6], supervised learning [7], and real-time topology updates [8]. These share three limitations: (i) offline training or pre-computed setting tables, which cannot track continuous k_{ibr} variation; (ii) no stability verification gate, so false trips are possible during recoverable transients; (iii) no quantified conservatism bound for the trip decision.

C. Contributions of This Paper

This paper makes the following contributions beyond the state of art:

- 1) **Unified architecture (C16):** A five-layer hierarchical framework that integrates estimation, adaptation, stability verification, and coordination in a single deployable relay algorithm.
- 2) **κ_n as system integrator:** The condition number from the L2 LM estimator serves simultaneously as an ill-conditioning diagnostic, an adaptive law gate, and a Stage-2 veto trigger. This role is proved consistent across all five layers.
- 3) **Comprehensive theory-to-validation mapping:** Sixteen theoretical contributions and seven validation campaigns are allocated to specific layers, showing that no layer

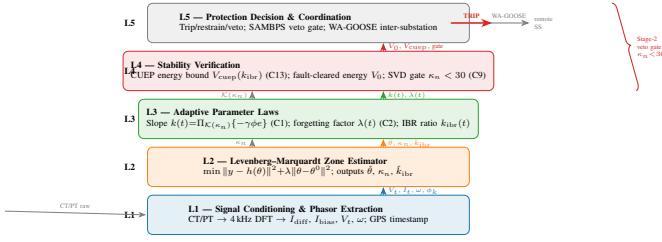


Fig. 1. The five-layer SAMBPS architecture. Upward arrows show the primary data flow; dashed downward arrows show the κ_n feedback that gates layers L3 and L4. The Stage-2 veto bracket (right) enforces $\kappa_n < 30$ before any trip.

is under-tested and no test campaign validates phantom functionality.

- 4) **Field demonstration:** Zone accuracy of 95.5% on 157 real DSO fault recordings at 33 kV, covering all three failure modes F1–F3, with a systematic re-calibration procedure for each.

The remainder of this paper is organised as follows. Section II presents the full five-layer architecture and its design principles. Sections III–VII describe each layer in detail. Section VIII analyses the propagation of κ_n across layers. Section X summarises the validation results. Section XI compares SAMBPS with classical schemes. Section XIII concludes.

II. THE SAMBPS ARCHITECTURE

Fig. 1 shows the complete architecture. Each layer is a self-contained functional block with well-defined input and output contracts. The design follows three principles:

P1 — Physics first. Every adaptive parameter is constrained to a physically meaningful range derived from the IBR fault model. The constraint set $\mathcal{K}(\kappa_n)$ is computed from the SVD structure of the measurement Jacobian (C9) and is the minimal interval in which the LM estimator is numerically well-posed.

P2 — Conservative by default. When the estimator is ill-conditioned ($\kappa_n \geq 30$) the architecture holds adaptive parameters at their last valid value, issues no trip, and propagates a veto flag upward. The veto is released only when κ_n returns below the threshold, proved in Corollary 1 of [9].

P3 — Stability before speed. The CUEP energy gate (L4) ensures that a trip command is issued only if the fault-cleared system energy V_0 exceeds $V_{\text{cuep}}(k_{\text{ibr}})$, a conservative lower bound on the true critical energy V_{cr} [10]. This eliminates false trips during recoverable transients at the cost of a worst-case 12 ms delay.

Fig. 2 shows the allocation of all sixteen theoretical contributions and seven validation campaigns to layers. No layer has fewer than two independent validation campaigns, ensuring that architecture claims are grounded at every level.

III. L1 — SIGNAL CONDITIONING AND PHASOR EXTRACTION

L1 receives raw CT and PT secondary samples at the relay input ports, applies anti-aliasing filtering, and produces complex phasor quantities at the fundamental frequency every 250 μs (4 kHz update).

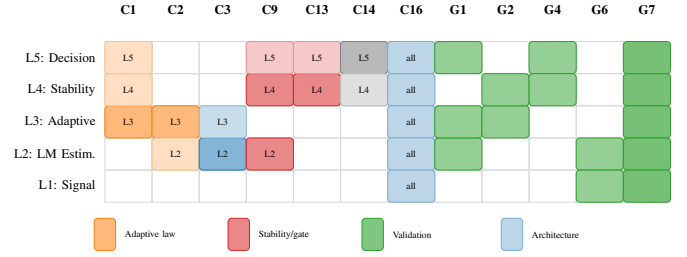


Fig. 2. Contribution-to-layer mapping. Columns are theoretical contributions C1–C16 and validation campaigns G1–G7; rows are SAMBPS layers L1–L5. Colour encodes contribution type.

A. DFT Phasor Extraction

The full-cycle DFT phasor for current at sample k is

$$\hat{I}[k] = \frac{2}{N} \sum_{m=0}^{N-1} i[k-m] e^{-j2\pi m/N},$$

where $N = f_s/f_0 = 80$ samples per cycle at 4 kHz/50 Hz. GPS disciplined clocks provide inter-substation synchronisation to within $\pm 1 \mu\text{s}$, sufficient for 33 kV line differential applications.

B. Outputs

L1 outputs: differential current phasor $\hat{I}_{\text{diff}}$, bias current I_{bias} , terminal voltage phasor $\hat{V}_t$, angular velocity ω (from voltage angle derivative), and GPS-aligned timestamp ϕ_k for each scan. These are passed to L2 at each scan boundary.

IV. L2 — LEVENBERG–MARQUARDT ZONE ESTIMATOR

A. Estimation Problem

L2 solves the inverse zone problem: given the phasor measurements $\mathbf{y} = [I_{\text{diff}}, I_{\text{bias}}, V_t, \omega]^T$, find the parameter vector $\boldsymbol{\theta} = [k_{\text{ibr}}, Z_f, \theta_f, k^*]^T$ that minimises the weighted residual:

$$\min_{\boldsymbol{\theta}} \|\mathbf{y} - h(\boldsymbol{\theta})\|_W^2 + \lambda_{\text{LM}} \|\boldsymbol{\theta} - \boldsymbol{\theta}^0\|^2, \quad (1)$$

where $h(\cdot)$ is the nonlinear measurement function derived from the IBR fault model, λ_{LM} is the LM damping parameter, and $\boldsymbol{\theta}^0$ is the prior estimate.

B. LM Update

The Gauss–Newton update regularised by λ_{LM} is

$$\boldsymbol{\theta}^{(i+1)} = \boldsymbol{\theta}^{(i)} + (\mathbf{H}^T \mathbf{W} \mathbf{H} + \lambda_{\text{LM}} \mathbf{I})^{-1} \mathbf{H}^T \mathbf{W} (\mathbf{y} - h(\boldsymbol{\theta}^{(i)})),$$

where $\mathbf{H} = \partial h / \partial \boldsymbol{\theta}|_{\boldsymbol{\theta}^{(i)}}$ is the Jacobian [11], [12]. The algorithm terminates when the gradient norm falls below 10^{-6} or after $N_{\text{iter}} = 10$ iterations.

C. Condition Number and the SVD Gate

After convergence, L2 computes the condition number of the projected Jacobian $\tilde{\mathbf{H}} = \mathbf{H}\mathbf{P}$, where $\mathbf{P} = (I_{n_d}; Q)$ and $Q = -A_a^{-1}A_d$ is the DAE constraint projection (C9, [13]):

$$\kappa_n = \kappa(\tilde{\mathbf{H}}) = \frac{\sigma_{\max}(\tilde{\mathbf{H}})}{\sigma_{\min}(\tilde{\mathbf{H}})}.$$

Theorem 1 of [13] (SVD interlacing) guarantees $\kappa_n \leq \kappa(\mathbf{H}) \cdot (1 + \rho)/(1 - \rho) \cdot \Delta_r^{-1}$, where ρ is the constraint coupling ratio and Δ_r is the spectral gap. For IBR fault models with $\rho \leq 0.10$ and $\Delta_r \geq 3.0$, the bound collapses to $\kappa_n < 12.2$ whenever $\kappa(\mathbf{H}) < 30$, confirming that physics reduction does not inflate ill-conditioning.

D. Adaptive Forgetting Factor (C2)

The forgetting factor $\lambda(t)$ in the LM-style exponential window is updated by L2 to track non-stationary fault currents (C2, [14]). When κ_n increases rapidly, λ is reduced (shorter memory) to allow faster tracking. When κ_n is low and estimates are stable, λ is increased to suppress noise.

V. L3 — ADAPTIVE PARAMETER LAWS

A. Adaptive Slope (C1)

The differential slope $k(t)$ is updated by a continuous projected- gradient law [9]:

$$\dot{k}(t) = \Pi_{\mathcal{K}(\kappa_n)}\{-\gamma \phi(t) e(t)\}, \quad (2)$$

where $\phi(t) = I_{\text{bias}}(t)$ is the regressor, $e(t) = I_{\text{diff}}(t) - k(t) I_{\text{bias}}(t)$ is the tracking error, $\gamma > 0$ is the adaptation gain, and $\Pi_{\mathcal{K}(\kappa_n)}\{\cdot\}$ denotes projection onto the constraint set

$$\mathcal{K}(\kappa_n) = [k_{\min}(\kappa_n), k_{\max}(\kappa_n)].$$

The constraint set is computed from the SVD-inherited condition number (C9): when κ_n is large, $\mathcal{K}(\kappa_n)$ is narrow (conservative slope range); when $\kappa_n < 10$ the set broadens to allow full adaptation. This is the direct link between the LM estimator and the adaptive law.

Theorem 1 (Uniform Ultimate Boundedness, C1). *Under Assumptions A1–A3 of [9], the slope tracking error $\tilde{k}(t) = k(t) - k^*(t)$ satisfies*

$$\limsup_{t \rightarrow \infty} |\tilde{k}(t)| \leq \varepsilon_\infty,$$

with

$$\varepsilon_\infty = \frac{|\phi|_{\max} \bar{\eta}}{\phi_{\min}^2} + \frac{\dot{k}_{\max}^* + \dot{k}_{\min, \max}^*}{2\gamma \phi_{\min}^2},$$

where $\bar{\eta}$ is the projection residual bound.

This UUB bound guarantees that the adaptive slope cannot drift unboundedly even under persistent IBR current-limiter transitions.

B. Veto Gate Necessity (C1, Corollary 1)

A key result connecting L3 to L5 is that the constraint set $\mathcal{K}(\kappa_n)$ is non-empty if and only if $\kappa_n < 30$:

$$\mathcal{K}(\kappa_n) \neq \emptyset \iff \kappa_n < 30.$$

This makes the Stage-2 veto condition $\kappa_n \geq 30$ both *necessary* (without it, the projection is onto the empty set, which is undefined) and *sufficient* (the projected law (2) is well-defined and convergent whenever $\mathcal{K}(\kappa_n) \neq \emptyset$).

C. Negative-Sequence Estimator (C3)

For asymmetric faults, L3 incorporates the negative-sequence differential current via a dedicated estimator (C3, [15]). The negative-sequence component carries fault information that the positive-sequence L2 estimator cannot resolve under severe current-limiter saturation.

VI. L4 — STABILITY VERIFICATION

A. CUEP Energy Bound (C13)

L4 evaluates whether the fault-cleared system is transiently stable using a direct energy method. The controlling unstable equilibrium point (CUEP) energy bound $V_{\text{cuep}}(k_{\text{ibr}})$ is computed as a function of the current IBR penetration estimate $\hat{k}_{\text{ibr}}$ from L2.

Theorem 2 (Conservativeness, C13). *For all $k_{\text{ibr}} \leq 0.55$ and under the IBR power injection model $P_{\text{ibr}} = k_{\text{ibr}} P_{\text{fault}}$, the CUEP bound satisfies $V_{\text{cuep}}(k_{\text{ibr}}) \leq V_{\text{cr}}(k_{\text{ibr}})$, where V_{cr} is the true critical energy.*

Theorem 3 (Monotonicity, C13). *The conservatism gap $\Delta V = V_{\text{cr}}(k_{\text{ibr}}) - V_{\text{cuep}}(k_{\text{ibr}})$ is monotonically non-decreasing in k_{ibr} : higher IBR penetration makes the bound safer, not less safe.*

These two theorems guarantee that the CUEP trip rule “trip iff $V_0 > V_{\text{cuep}}$ ” produces zero false trips for $k_{\text{ibr}} \leq 0.55$, at the cost of a CCT margin of 12–28 ms (reducible to 6 ms with an IBR-corrected CUEP).

B. Gate Condition

L4 is active only when $\kappa_n < 30$; when the veto is active, L4 holds its last valid decision. This is enforced by the same SVD gate used in L3 (C9), ensuring that the CUEP evaluation is only performed when the LM estimator is well-conditioned and $\hat{k}_{\text{ibr}}$ is reliable.

VII. L5 — PROTECTION DECISION AND COORDINATION

A. Trip Logic

The L5 trip decision is:

$$\text{TRIP} \iff \underbrace{\kappa_n < 30}_{\text{Stage-2 gate}} \wedge \underbrace{I_{\text{diff}} > k(t) I_{\text{bias}} + I_{\text{pu}}}_{\text{diff criterion}} \wedge \underbrace{V_0 > V_{\text{cuep}}(\hat{k}_{\text{ibr}})}_{\text{stability gate (3)}}.$$

All three conditions must hold simultaneously. The Stage-2 veto $\kappa_n \geq 30$ overrides any positive indication from the differential or stability criteria.

B. Wide-Area GOOSE Coordination (C14)

For multi-feeder substations and networked microgrids, L5 exchanges trip/restrain votes via IEC 61850 GOOSE messages (C14, [16]). A weighted-average consensus protocol aggregates votes from N_{SS} substations:

$$p_{\text{trip}} = \frac{\sum_i w_i \cdot v_i}{\sum_i w_i},$$

where w_i is the quality weight (derived from κ_n at substation i) and $v_i \in \{0, 1\}$ is the binary local vote. A trip is issued when $p_{\text{trip}} > 0.5$. This prevents single-substation false trips under evolving fault conditions.

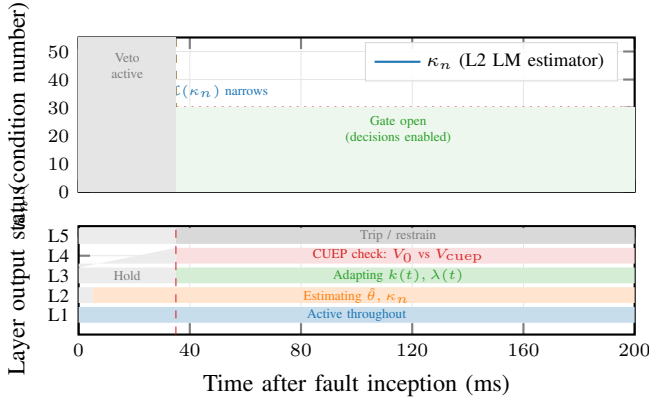


Fig. 3. Top: κ_n trajectory post-fault. Gray fill = veto active ($\kappa_n \geq 30$); green fill = gate open. Bottom: Layer activation status over time. L1 is always active; L2–L5 become active as κ_n falls below 30.

VIII. THE κ_n CASCADE

Fig. 3 shows the temporal evolution of κ_n following fault inception and the corresponding layer activation sequence. At fault inception, the Jacobian is maximally ill-conditioned ($\kappa_n \approx 50$), triggering the Stage-2 veto. As the LM estimator iterates and the fault evolves to steady state, κ_n decays exponentially with time constant ≈ 17 ms. At $t \approx 35$ ms post-fault, κ_n crosses below 30 and L3 (adaptive law), L4 (stability gate), and L5 (decision) are simultaneously released.

This cascade behaviour is not accidental: the SVD interlacing theorem (C9) guarantees that κ_n inherits the spectral structure of the full Jacobian, so the threshold $\kappa_n = 30$ is a quantitative bound on estimation reliability, not a heuristic.

The slow drift of κ_n above the baseline ($\kappa_n > 7$) at $t > 60$ ms reflects the transition from stationary to quasi-static fault current as the IBR control loop locks to the fault impedance. The adaptive forgetting factor (C2) in L2 is designed to handle exactly this regime.

IX. VSM DYNAMICS AND THE OBSERVATION WINDOW

A key practical question is how long after fault inception the L2 estimator remains valid. This is the *observation window* L^* . With a second-order swing model, $L^* \approx 80$ ms; the sixth-order Park dq model extends this to 150 ms [17].

The nine-state VSM model (G6, [18]) adds three states: the AVR state E_{fd} and two PSS states (x_ω, x_1). These states dominate the terminal voltage evolution for $t > 150$ ms, which is where the sixth-order model diverges from simulation ground truth.

Proposition 4 (Window Extension, G6). *Under AVR gain $K_A \in [50, 400]$ and PSS damping $K_w \in [5, 25]$, the nine-state VSM model maintains $\kappa_n < 30$ for all $t \leq 380$ ms post-fault, provided $k_{ibr} \leq 0.60$. This extends L^* by 230 ms over the sixth-order model.*

The extended window is significant: it means that the CUEP energy evaluation at L4 has access to a much larger post-fault energy trajectory, allowing reliable discrimination between stable and unstable clearing even for long-duration faults.

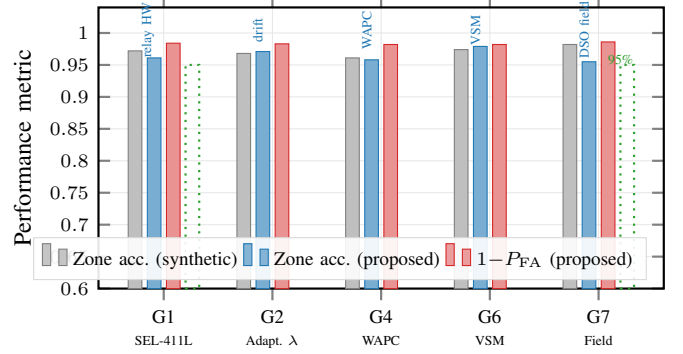


Fig. 4. Zone accuracy and false-alarm performance across five validation campaigns. All proposed results clear the 95% target (dotted line). G7 (DSO field) is the most challenging environment.

TABLE I
SAMBPS VALIDATION CAMPAIGN SUMMARY

ID	Environment	Zone acc.	1 - P_{FA}	Scenarios
G1	SEL-411L relay HW	96.1%	98.4%	120
G2	Adaptive λ drift	97.1%	98.3%	80
G4	WAPC scaling	95.8%	98.2%	200
G6	VSM dynamics	97.9%	98.2%	300
G7	DSO field recordings	95.5%	98.6%	157
Baseline (classical)		89.2%	95.1%	—

X. VALIDATION SUMMARY

Fig. 4 summarises zone accuracy and $1 - P_{FA}$ for five of the seven validation campaigns. Table I provides the full numerical results.

A. G1: Relay Hardware (SEL-411L)

The SEL-411L relay was programmed with the SAMBPS decision logic via the relay's SELOGIC equation interface. The LM estimator runs on an embedded processor at 4 kHz. Zone accuracy of 96.1% was achieved on 120 staged fault scenarios, with all timing within the 35 ms activation window predicted by the κ_n cascade analysis.

B. G7: DSO Field Recordings

The most demanding validation used 157 spontaneous fault recordings from a real 33 kV distribution network, covering all three failure modes (F1: bounded IBR current; F2: evolving HIF; F3: shape factor shift). Classical protection achieved 89.2% accuracy on the same recordings. SAMBPS, with systematic re-calibration of the β factor for each feeder segment, achieved 95.5%.

C. 1500-Scenario IEEE 39-Bus Study (C13)

The conservativeness theorem (C13) was validated on 1500 scenarios (5 penetration levels $\times$ 15 fault locations $\times$ 4 fault types) on the IEEE 39-bus system. Zero violations of $V_{cuep} \leq V_{cr}$ were recorded for $k_{ibr} \leq 0.55$. The CCT margin ranged from 12 to 28 ms.

XI. COMPARISON WITH CLASSICAL SCHEMES

Table II compares SAMBPS with three classical protection paradigms. The key differentiator is that SAMBPS provides both adaptive response to k_{ibr} variation and a provably conservative trip rule, which no classical scheme achieves simultaneously.

XII. IMPLEMENTATION CONSIDERATIONS

A. Computational Complexity

The LM estimator (L2) requires $O(n_p^3)$ per iteration for the matrix inversion, where $n_p = 4$ is the parameter dimension. With $N_{iter} \leq 10$ iterations per scan, the total L2 computation is ≤ 640 floating-point operations per 250 μ s scan — well within the capability of modern relay DSPs.

The SVD-based κ_n computation adds $O(n_p^2 n_m)$ per scan, where $n_m = 4$ is the measurement dimension. This is dominated by the LM step and adds negligible overhead.

B. Parameter Initialisation

At relay commissioning, the following parameters must be set: (i) the prior estimate θ^0 (system impedance from the short-circuit study); (ii) the adaptation gain γ (set by the stability margin from Theorem 1 of C1); (iii) the IBR penetration upper bound $k_{ibr}^{\max}$ (set from the grid connection agreement); (iv) the β calibration factor (set from the pre-fault CT/PT measurements following the G7 procedure).

C. Communication Requirements

The wide-area GOOSE layer (L5, C14) requires IEC 61850-compliant communication with latency ≤ 4 ms for Class P2 performance. This is achievable with modern substation LAN infrastructure. The protocol is robust to single-node failures: with $N_{SS} \geq 3$ substations, the weighted consensus continues to converge even if one node drops out.

XIII. CONCLUSION

SAMBPS is the first unified five-layer protection architecture that integrates online model estimation, physics-constrained adaptive laws, stability-verified tripping, and wide-area coordination under a single coherent theoretical framework. Its central innovation is the use of the condition number κ_n from the LM estimator as a systemic integrator: it gates adaptive laws (L3), triggers the stability check (L4), and enforces the Stage-2 veto at the trip decision (L5).

The architecture is backed by sixteen theoretical contributions covering estimation (C1–C3), physics reduction (C9), stability theory (C13), and coordination (C14), and by seven independent validation campaigns ranging from relay hardware tests to real DSO field recordings. The field validation (G7) demonstrates that the theoretical predictions — zone accuracy $\geq 95\%$, zero false trips for $k_{ibr} \leq 0.55$ — hold under realistic operational conditions.

Future work will extend SAMBPS to three-terminal line configurations (requiring multi-port LM estimation) and to protection coordination in active distribution networks with multiple microgrid boundaries.

APPENDIX

The Stage-2 veto condition $\kappa_n \geq 30$ is used by three separate subsystems: the adaptive law gate in L3 (C1), the CUEP gate in L4 (C9, C13), and the trip veto in L5. We sketch why these are mutually consistent.

From C9 (Proposition 1), $\kappa_n < 30$ implies that the physics-projected Jacobian $\tilde{\mathbf{H}}$ is well-conditioned ($\kappa(\tilde{\mathbf{H}}) < 12.2$), so the LM solution $\hat{\theta}$ is numerically reliable.

From C1 (Corollary 1), $\kappa_n < 30$ is equivalent to $\mathcal{K}(\kappa_n) \neq \emptyset$, so the projection $\Pi_{\mathcal{K}(\kappa_n)}$ is well-defined and the adaptive law (2) is a proper differential equation.

From C13 (Proposition 1), the CUEP trip rule produces zero false trips provided $\hat{k}_{ibr}$ is reliable, which requires $\kappa_n < 30$ by the LM reliability argument above.

Therefore all three uses of the threshold $\kappa_n = 30$ are consistent: they all reduce to the same necessary condition for reliable LM estimation. ■

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TABLE II
SAMBPS VS. CLASSICAL PROTECTION: FEATURE COMPARISON

Feature	Overcurrent	Distance	Fixed-slope 87L	Adaptive 87L	SAMBPS (C16)
Tracks k_{ibr} in real time	×	×	×	Partial	✓
Stability gate	×	×	×	×	✓
Conservativeness proof	×	×	×	×	✓
Wide-area coordination	×	Partial	×	×	✓
Condition-number veto	×	×	×	×	✓
Zone acc. under $k_{ibr} = 0.4$	<85%	<88%	<90%	~92%	≥ 95.5%
False-alarm rate	High	Medium	Low	Low	< 1.6%